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Endogenous Treatment Models with Social Interactions: An Application to the Impact of Exercise on Self-Esteem

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1 Big picture

- **Question.** How can we estimate **endogenous treatment models** when both the treatment decision and the outcome are shaped by **social interactions**, and what do treatment effects mean once one individual's treatment can affect others through a network?
- **Setting.** The paper studies a large-network environment in which individuals make simultaneous treatment decisions and outcomes are also socially interdependent. The empirical application examines the impact of **exercise** on **self-esteem** among US adolescents.
- **Core challenge.** In standard endogenous treatment models, the treatment decision is endogenous because unobservables in the treatment and outcome equations are correlated. Here there is an additional difficulty: peers' expected treatment and expected outcomes enter both equations, so there are **general equilibrium network effects** on top of ordinary treatment endogeneity.
- **Main methodological contribution.**
 1. The paper builds an endogenous treatment model with **peer effects in both the treatment and the outcome equations**.
 2. It represents these interactions through two **discrete Bayesian games** under incomplete information.
 3. It proposes a **nested pseudo joint likelihood (NPJL)** estimator that is computationally tractable in large networks.
 4. It defines treatment effects that distinguish between **partial effects** and **composite equilibrium effects** that propagate through the network.
- **Why this matters.** If policy changes one person's treatment decision, the effect need not stop there. Friends may change their treatment decisions, and outcomes may respond through both treatment and outcome networks. So standard treatment-effect objects can be badly incomplete.
- **Main empirical question.** What is the effect of frequent exercise on self-esteem once we allow for both **endogeneity of exercise** and **peer effects in exercise and self-esteem**?
- **Main empirical results.**
 - An individual's exercise choice responds positively to the **expected exercise of friends**.
 - After accounting for endogeneity, exercise has a **large positive and statistically significant** effect on self-esteem.
 - The estimated cross-equation error correlation is strongly negative in the baseline specification, indicating that unobserved factors were masking the positive effect of exercise.
 - Peer effects in the self-esteem equation are weak at the median but appear **at low self-esteem levels**.
 - Counterfactual interventions aimed at selected students generate sizable **spillover effects** on the rest of the network.
 - Targeting the most popular students is slightly more effective than targeting the poorest students when the objective is to maximize total outcome effects.

- **Economic takeaway.** In a networked treatment environment, the relevant policy object is not just the direct effect of treatment on the treated individual. What matters is the **equilibrium multiplier** created by peer effects in both treatment and outcomes.

2 Data and measurement (key objects)

2.1 Data source and sample

- **Main dataset.** Wave I of the **National Longitudinal Study of Adolescent Health (Add Health)**.
- **Timing.** The survey was conducted in **1994–95**.
- **Population.** Students in grades **7–12** from a sample of representative schools.
- **Sample size.** The usable estimation sample contains **9,036 students**.

2.2 Network, treatment, outcome, and covariates

- **Network.** Each student lists up to **five best female friends** and **five best male friends**. The friendship network is constructed from these nominations.
- **Treatment variable: exercise.** Students answer:

“How many times in a normal week do you work, play, or exercise hard enough to make you sweat and breathe heavily?”

The paper defines the treatment indicator as

$$\text{Exercise}_i = 1$$

if the student reports exercising **six or more times per week**, and 0 otherwise.

- **Outcome variable: self-esteem.** Self-esteem is built from six Rosenberg-style items:

1. you have a lot of good qualities,
2. you have a lot to be proud of,
3. you like yourself just the way you are,
4. you feel like you are doing everything just about right,
5. you feel socially accepted,
6. you feel loved and wanted.

Responses are coded from 0 to 4 and summed into an index ranging from **0 to 24**.

- **Binary outcome construction.** Because the structural model is binary, the paper converts the self-esteem index into indicators for being above different thresholds, especially the **median** and various **quantiles/deciles**.
- **Observed covariates.** The empirical specification includes age, gender, GPA, self-reported intelligence, race (white), and family income.

2.3 Key descriptive objects

- **Age.** Mean age is **15.638** years.
- **Gender composition.** **52.5%** of the sample is female.
- **Race.** **61.9%** of the sample identifies as white.
- **Exercise prevalence.** The exercise treatment indicator has a mean of **0.408**, so about **40%** of the sample exercises at least six times per week.
- **Self-esteem index distribution.** The raw self-esteem index has mean close to **19**; the first quartile is **17**, the median is **18**, and the third quartile is **22**.
- **Binary self-esteem indicator.** In the median-threshold specification, the self-esteem outcome has mean **0.497**.
- **Intelligence categories.** The self-reported intelligence indicators have shares **0.048** for categories 1–2, **0.589** for categories 3–4, and **0.363** for categories 5–6.

2.4 Unobservables, information sets, and timing

- **Private unobservables.** The outcome and treatment equations contain errors u_i and v_i , which are private information known only to individual i .
- **Public information sets.**

$$I_i^O = \{X_i, Z_i, D_i, F_i\}_{i=1}^n,$$
$$I_i^T = \{Z_i, F_i\}_{i=1}^n.$$

- **Interpretation.** Individuals do *not* observe realized peer choices when making their own decisions; they respond to **beliefs about peers' choices and outcomes**.
- **Sparse-network condition.** The number of friends is uniformly bounded, which keeps the network sparse as the sample grows and helps with inference.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline endogenous treatment model with peer effects

The core econometric model is

$$Y_i = 1\{X_i'\beta^O + \gamma D_i + dW_i^* + u_i > 0\}, \quad (1)$$

$$D_i = 1\{Z_i'\beta^T + aV_i^* + v_i > 0\}. \quad (2)$$

- Y_i : binary outcome (self-esteem above a threshold).

- D_i : binary treatment (high-frequency exercise).
- γ : direct effect of treatment on the outcome.
- d : peer effect in the outcome equation.
- a : peer effect in the treatment equation.
- (u_i, v_i) : potentially correlated unobservables.

Interpretation. The model combines ordinary endogenous treatment logic with social interactions. Treatment affects the outcome directly through γ , while peer beliefs shift treatment and outcomes through V_i^* and W_i^* .

3.2 Peer-effect objects and local averages

The social-interaction terms are constructed as

$$W_i^* = \frac{1}{N_i} \sum_{j \in F_i} \mathbb{E}(Y_j \mid I^O), \quad (3)$$

$$V_i^* = \frac{1}{N_i} \sum_{j \in F_i} \mathbb{E}(D_j \mid I^T). \quad (4)$$

- $F_{ij} = 1$ means that individual i considers j a friend.
- F_i is the set of friends of i , and N_i is the number of those friends.
- The model uses **local average expected peer behavior** rather than identities of peers.

Interpretation. The peer variables summarize expected treatment and expected outcomes of friends. This is a reduced but tractable way of embedding network interaction in a large game.

3.3 Incomplete information and Bayesian games

The treatment and outcome interactions are represented by two separate **Bayesian games**. Individuals move simultaneously and do not observe realized peer choices. Instead, they condition on beliefs such as $\mathbb{E}(D_j \mid I^T)$ and $\mathbb{E}(Y_j \mid I^O)$.

Interpretation. This incomplete-information structure is crucial. It makes the large-network problem tractable, allows equilibrium to be represented through conditional choice probabilities, and avoids treating peers' realized choices as endogenous regressors.

3.4 Conditional choice probabilities and equilibrium representation

Under the assumption that (u_i, v_i) are independent across individuals and jointly normal with unit variances and correlation ρ , the treatment CCP is

$$P_{Ti}^* = \mathbb{P}(D_i = 1 \mid I^T) = \Phi(Z_i' \beta^T + a V_i^*). \quad (5)$$

The outcome CCP is

$$P(Y_i = 1 \mid I^O) = D_i \frac{\Phi_2(X'_i \beta^O + \gamma D_i + dW_i^*, Z'_i \beta^T + aV_i^*; \rho)}{\Phi(Z'_i \beta^T + aV_i^*)} + (1 - D_i) \frac{\Phi(X'_i \beta^O + \gamma D_i + dW_i^*) - \Phi_2(X'_i \beta^O + \gamma D_i + dW_i^*, Z'_i \beta^T + aV_i^*; \rho)}{1 - \Phi(Z'_i \beta^T + aV_i^*)}. \quad (6)$$

The best-response mappings are

$$\Psi_{Ti}(v; P_T) = \Phi(Z'_i \beta^T + aV_i(P_T)), \quad (7)$$

with an analogous mapping $\Psi_{Oi}(v; P_O)$ for the outcome equation.

The paper assumes:

- a bounded number of friend nominations,
- **moderate social influence**, specifically

$$0 \leq a < \sqrt{2\pi}, \quad |d| < \frac{1}{\sup w_{Oi}(\cdot)},$$

where $w_{Oi} = \partial \Psi_{Oi} / \partial W_i$.

Interpretation. These restrictions make the best-response mappings contractive. Hence the two Bayesian games have a **unique BNE** in CCP space.

3.5 Treatment effects: partial and composite

The paper distinguishes two treatment-effect concepts.

Partial treatment effect

A partial treatment effect changes the treatment status of a focal individual while **holding everyone else's treatment status fixed**. For individual i ,

$$PTE_i = P(Y_i = 1 \mid D_i = 1, I^O) - P(Y_i = 1 \mid D_i = 0, I^O). \quad (8)$$

Averaging gives

$$APTE = \frac{1}{n} \sum_{i=1}^n PTE_i. \quad (9)$$

Interpretation. This captures the direct impact of switching an individual's treatment status, plus outcome-game feedback, but not full treatment-game general equilibrium.

Composite treatment effect

A composite treatment effect changes the **index in the treatment CCP**, so the equilibrium treatment probabilities themselves move from P_T^0 to P_T^1 . This can alter the treatment choices and

outcomes of the focal person, their friends, their friends' friends, and so on. The paper simulates the unobserved shocks to compute this effect.

Interpretation. The composite effect contains three channels:

1. the direct effect of own treatment through γ ,
2. an indirect effect through the treatment peer term V_i^* ,
3. an indirect effect through the outcome peer term W_i^* .

This is the right object for policy analysis in networks.

3.6 NPJL estimator

Let $P[n]$ collect the treatment and outcome CCPs. The paper defines the pseudo-likelihood

$$L_n(v; P[n]) = \frac{1}{n} \sum_{i=1}^n l_i(v; P[n]), \quad (10)$$

where $l_i(v; P[n])$ is the pseudo log conditional probability of observing (Y_i, D_i) .

The **nested pseudo joint likelihood** estimator proceeds iteratively:

1. start from conjectured treatment and outcome CCPs,
2. given these CCPs, estimate the structural parameters by bivariate probit,
3. update CCPs using the estimated best-response mappings,
4. iterate until the CCPs converge.

The update step is

$$P^{(K+1)} = \Psi \left(\hat{v}^{(K+1)}; P^{(K)} \right). \quad (11)$$

Interpretation. NPJL avoids trying to solve the full high-dimensional problem in one step. Instead, it alternates between parameter estimation and equilibrium updating.

3.7 Asymptotic results

Under full-rank and regularity conditions, the estimator is consistent and asymptotically normal:

$$\hat{v} \xrightarrow{P} v_0, \quad (12)$$

$$\sqrt{n}(\hat{v} - v_0) \xrightarrow{d} N \left(0, \Omega_1(v_0) \Omega_2^{-1}(v_0) \Omega_1'(v_0) \right). \quad (13)$$

In addition, the estimated equilibrium CCPs satisfy

$$\|\hat{P}[n] - P^*[n]\| = O_P \left(\frac{1}{n} \right). \quad (14)$$

Interpretation. The iterative equilibrium estimator is not just computationally convenient; it is also formally justified.

3.8 Extension to sample-selection models

The procedure also extends to selection models motivated by Heckman, but only in the special case with **no peer effects in the outcome equation**. This restriction matters because once we move from the full sample to the selected sample, peers may disappear from the observed network. Without peer effects in the outcome equation, the model can accommodate continuous selected outcomes such as wages.

Interpretation. The framework is broader than the exercise application. It can be used in treatment and selection settings, but the outcome-network part is what prevents a direct continuous-outcome extension in the large-network case.

3.9 Empirical specification for exercise and self-esteem

The empirical model is

$$\text{Self-Esteem}_i = 1\{X_i'\beta^O + \gamma \text{Exercise}_i + dW_i^* + u_i > 0\}, \quad (15)$$

$$\text{Exercise}_i = 1\{X_i'\beta^T + aV_i^* + v_i > 0\}. \quad (16)$$

The baseline uses the same observed covariates in both equations and relies on the peer-effects term in the treatment equation for identification.

Interpretation. The model asks whether exercise raises self-esteem once we correctly model (i) endogenous exercise choice and (ii) peer dependence in both exercise and self-esteem.

4 Identification and instruments (what makes the estimates credible)

4.1 What standard approaches miss

- Standard endogenous treatment models typically allow correlated unobservables across treatment and outcome equations but omit peer effects.
- Standard peer-effects models often treat treatment as exogenous or do not allow peer effects simultaneously in treatment and outcome.
- If peer effects are omitted, the control-function or joint-likelihood correction for endogeneity is generally misspecified.

4.2 What identifies the model

- The model is identified through its **parametric structure**, the **bivariate normal dependence** between u_i and v_i , and the inclusion of **peer effects in the treatment equation**.
- The paper explicitly notes that the same exogenous covariates can appear in both equations because identification comes from the network structure and the treatment peer term, not from a conventional exclusion restriction alone.

- Full-rank conditions on the augmented covariate vectors in treatment and outcome equations ensure local identification.

4.3 What the private-information assumption buys

- Individuals respond to **expected** peer treatment and outcome, not realized peer choices.
- This means W_i^* and V_i^* can be treated as exogenous objects even though the treatment itself is endogenous.
- It also makes the large-network Bayesian game tractable and supports contraction-based uniqueness.

Key intuition. If peers' realized choices entered directly, they would inherit additional endogeneity from common unobservables across network members. The incomplete-information structure avoids that problem.

4.4 Empirical credibility of the application

- The exercise threshold (six or more times per week) is chosen to isolate activity likely to include exercise beyond school-time routine behavior.
- For adolescents, it is plausible that they know only rough expectations about friends' exercise and self-esteem, rather than exact realizations.
- The sample contains a rich friendship network and a large enough population to study targeted interventions.

4.5 Important limitations

- The paper does **not** allow for publicly shared unobserved shocks that influence multiple individuals simultaneously.
- Friendship networks are assumed **exogenous**.
- The approach is **parametric**; relaxing those assumptions would make estimation much harder.
- The binary treatment/binary outcome structure is a tractability device. Continuous variables are handled by thresholding, not by estimating a continuous-outcome network game directly.

4.6 Relation to the literature

- Connects the endogenous treatment / Roy-model tradition to the social-interactions literature.
- Uses tools from incomplete-information discrete games and CCP-based fixed-point estimation.
- Extends earlier social-interaction binary-choice models by adding endogenous treatment and peer effects in the outcome equation.

5 Tables and figures

5.1 Figure 1: peer effects and treatment effects

Read:

- The figure shows three individuals with peer links across both treatment choices (D_1, D_2, D_3) and outcomes (Y_1, Y_2, Y_3).
- Direct connections run through friendship links, but effects can travel further through friends of friends.
- The figure is there to emphasize that treatment effects in a network are **system objects**, not isolated individual objects.

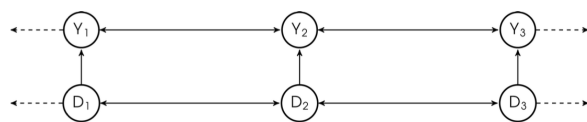


FIG. 1.—Peer effects and treatment effects.

TABLE 1
STATISTIC SUMMARY OF VARIABLES

Variable	Mean	Standard Deviation
Age	15.638	1.635
Female	.525	.518
GPA	2.834	.797
Intelligence (1 and 2)	.048	.214
Intelligence (3 and 4)	.589	.492
Intelligence (5 and 6)	.363	.481
White	.619	.486
Family income	.049	.055
Exercise	.408	.491
Self-esteem	.497	.500

5.2 Table 1: summary statistics

Read:

- The sample is young, roughly balanced by gender, and mostly white.
- Around **40%** of students satisfy the high-exercise treatment definition.
- The self-esteem distribution is relatively high on average but still leaves meaningful mass in the lower tail, which matters later when the paper studies low-self-esteem thresholds.

5.3 Table 2: main NPJL estimates

Read:

- In the **exercise equation**, peer effects are positive and statistically significant:

$$\hat{a} = 0.157 \quad (0.061).$$

This is evidence that students exercise more when they expect their friends to exercise more.

- The paper reports that moving the peer-exercise variable from its minimum to its maximum raises the average probability of exercise from **0.396** to **0.440**.

- In the **self-esteem equation**, exercise has a large positive coefficient:

$$\hat{\gamma} = 1.296 \quad (0.195),$$

which is highly statistically significant.

- The baseline peer effect in the self-esteem equation is small and not statistically different from zero:

$$\hat{d} = -0.020 \quad (0.050).$$

- The estimated error correlation is strongly negative:

$$\hat{\rho} = -0.695,$$

with a profiled confidence interval approximately $[-0.894, -0.334]$.

Interpretation. Ignoring endogeneity would understate the positive effect of exercise on self-esteem because unobserved factors move exercise and self-esteem in opposite directions.

TABLE 2
NESTED PSEUDO JOINT LIKELIHOOD ESTIMATION OF EXERCISE
AND SELF-ESTEEM

Variable	Estimate	Standard Error
A. Exercise		
Age	-.106***	.009
Female	-.668***	.027
GPA	.049***	.018
Intelligence (3 and 4)	.024	.064
Intelligence (5 and 6)	.085	.067
White	.224***	.029
Family income	.659***	.250
Intercept	1.378***	.158
Peer effects	.157***	.061
B. Self-Esteem		
Age	.004	.015
Female	.088	.078
GPA	-.005	.018
Intelligence (3 and 4)	.143**	.063
Intelligence (5 and 6)	.475***	.081
White	-.171***	.030
Family income	-.233	.242
Exercise	1.296***	.195
Intercept	-.763**	.292
Peer effects	-.020	.050
ρ	-.695**	-.894 to -.334

NOTE.—For ρ , profiled confidence interval is reported.

** Statistically significant at the 5% level.

*** Statistically significant at the 1% level.

TABLE 3
NESTED PSEUDO JOINT LIKELIHOOD ESTIMATION OF EXERCISE
AND SELF-ESTEEM (BOTTOM 10%)

Variable	Estimate	Standard Error
A. Exercise		
Age	-.107***	.009
Female	-.665***	.027
GPA	.050***	.019
Intelligence (3 and 4)	.028	.065
Intelligence (5 and 6)	.091	.068
White	.229***	.029
Family income	.672***	.250
Intercept	1.372***	.159
Peer effects	.154**	.062
B. Self-Esteem		
Age	-.003	.019
Female	-.141	.108
GPA	.064**	.025
Intelligence (3 and 4)	.282***	.075
Intelligence (5 and 6)	.460***	.086
White	-.118***	.045
Family income	-.161	.352
Exercise	.830***	.321
Intercept	.536	.455
Peer effects	.075*	.041
ρ	-.251	-.491 to .017

* Statistically significant at the 10% level.

** Statistically significant at the 5% level.

*** Statistically significant at the 1% level.

5.4 Table 3: bottom 10% self-esteem specification

Read:

- When the outcome is whether self-esteem is above the **bottom decile**, exercise remains positive and significant:

$$\hat{\gamma} = 0.830 \quad (0.321).$$

- The striking difference is that the peer-effect coefficient in the self-esteem equation becomes positive and marginally significant:

$$\hat{d} = 0.075 \quad (0.041).$$

- This suggests **contagion in self-esteem at the lower tail** of the self-esteem distribution.

Interpretation. Peer effects in outcomes are not uniform across the distribution. They matter most for individuals with relatively low self-esteem.

5.5 Table 4 and Table 5: counterfactual interventions

TABLE 4
COUNTERFACTUAL DECISIONS AND OUTCOMES

	Observed	After Index Change
A. Counterfactual 1		
Exercise	357	538*** (+50.7%) (16.318)
High self-esteem	494	585*** (+18.4%) (15.053)
APTE		.626
B. Counterfactual 2		
Exercise	476	614*** (+29.0%) (15.131)
High self-esteem	529	614*** (+16.1%) (16.273)
APTE		.563

*** Statistically significant at the 1% level.

TABLE 5
COUNTERFACTUAL DECISIONS AND OUTCOMES FOR WHOLE SAMPLE

	Observed	Counterfactual 1	Counterfactual 2
Exercise	3,684	3,872*** (+5.1%) (44.286)	3,883*** (+5.4%) (41.646)
High self-esteem	4,489	4,600** (+2.5%) (48.852)	4,608*** (+2.7%) (45.301)

** Statistically significant at the 5% level.

*** Statistically significant at the 1% level.

Read:

- The paper studies two targeted policies, each implemented as a **0.5 increase** in the treatment index for a chosen group.
- **Counterfactual 1: 1,000 poorest students.**
 - Targeted-group exercise rises from **357** to **538**.
 - Targeted-group above-median self-esteem rises from **494** to **585**.
 - APTE for this group is **0.626**.
 - In the whole sample, exercise rises from **3,684** to **3,872**, and above-median self-esteem rises from **4,489** to **4,600**.
- **Counterfactual 2: 1,000 most popular students.**
 - Targeted-group exercise rises from **476** to **614**.
 - Targeted-group above-median self-esteem rises from **529** to **614**.
 - APTE for this group is **0.563**.
 - In the whole sample, exercise rises from **3,684** to **3,883**, and above-median self-esteem rises from **4,489** to **4,608**.

Why these tables matter.

- The total-sample responses exceed the targeted-group responses because the policy propagates through the network.
- For the popular-student intervention, the increase in exercise is **138** in the targeted group but **199** in the full sample, so **44%** of the exercise increase is spillover.
- The increase in high self-esteem is **85** in the targeted group but **119** in the full sample, so **27%** is spillover.
- The most popular students are the slightly more effective target if the planner wants to maximize the total number of affected individuals.

6 Short summary

- **Method.** Model treatment and outcome as two linked Bayesian games with peer effects, estimate them by NPJL, and compute both partial and composite treatment effects.
- **Identification.** Identification comes from the parametric joint structure, incomplete-information equilibrium representation, and peer effects in the treatment equation.
- **Application.** Exercise and self-esteem among 9,036 adolescents in Add Health with friendship-network data.
- **Main result.** Expected peer exercise raises individual exercise; once endogeneity is accounted for, exercise has a large positive effect on self-esteem; peer effects in self-esteem appear mainly at the lower tail; and targeted interventions generate meaningful network spillovers.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper provides a workable structural framework for endogenous treatment models with social interactions in both treatment and outcome equations.
- It shows that treatment effects in networks must account for equilibrium feedback through peers' treatment and peers' outcomes.
- In the application, exercise has a strong positive effect on self-esteem once one correctly handles endogeneity and network interaction.

7.2 Economic intuition

- A student's exercise decision depends partly on what they expect their friends to do.
- If one student's treatment status changes, this alters others' expected behavior and can move their treatment decisions too.

- Once those treatment choices shift, outcomes may move directly through treatment and indirectly through outcome contagion.
- As a result, the relevant policy effect is a **social multiplier**, not a single-person coefficient.

7.3 Takeaway

This paper is useful because it turns a hard conceptual point into an estimable empirical framework: in network settings, **endogeneity and peer effects cannot be studied separately**. The main empirical lesson is equally clear: exercise appears to improve self-esteem substantially, and policies that raise exercise can have broader benefits because they propagate through social networks.